

Transformer Neural Networks

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RECAP - Part 1

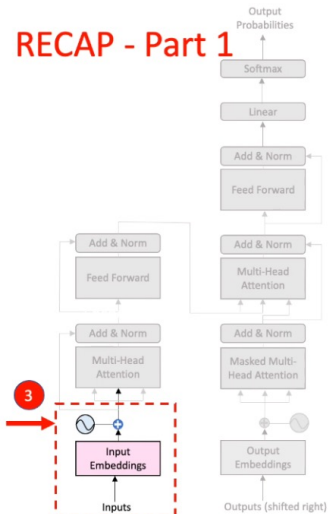
Dialogue Completer



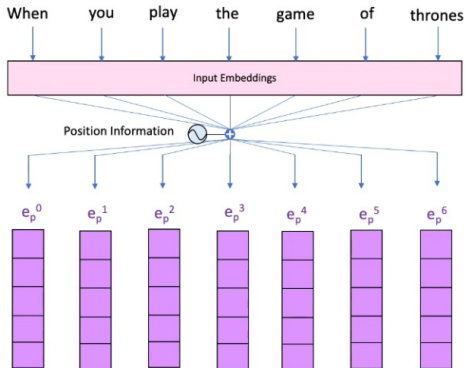
Input When you play the Game of Thrones ...

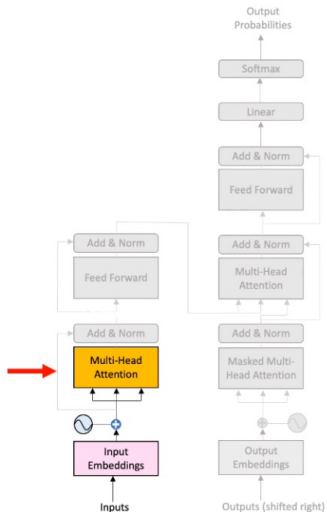
Transformer Output ?

RECAP - Part 1



Recap

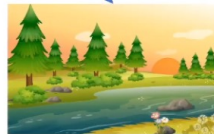




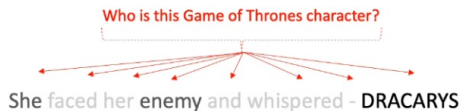
Multi-Head Attention

Self-Attention

He went to the bank and learned of his empty account, after which he went to a river bank and cried.



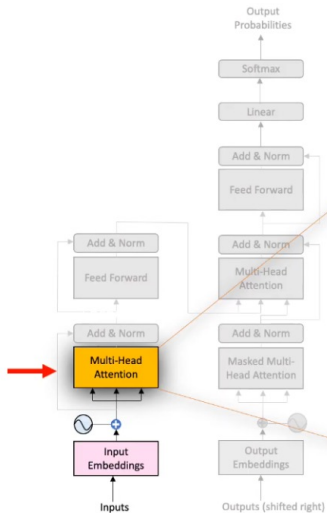
Simple-Attention



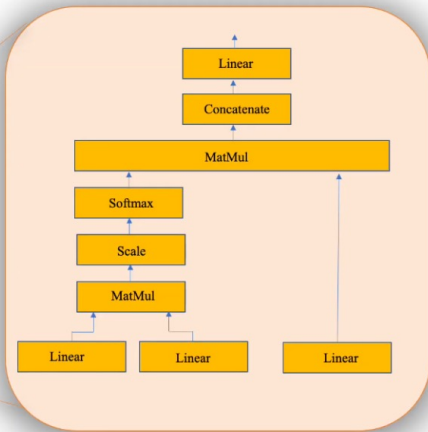
Self-Attention

He went to the bank and learned of his empty account, after
which he went to a river bank and cried.

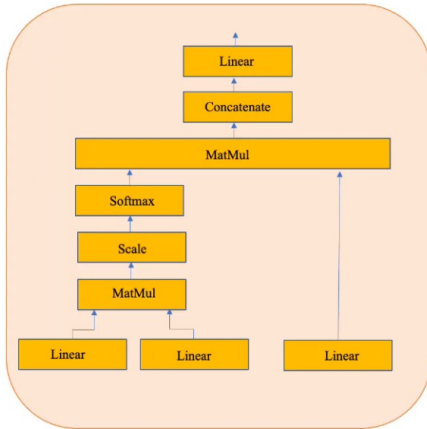
The diagram shows two sentences. The first sentence is "He went to the bank and learned of his empty account, after" and the second sentence is "which he went to a river bank and cried." A blue arc connects the word "bank" in the first sentence to the word "bank" in the second sentence, illustrating a self-attention mechanism where the model focuses on the same word in different parts of the input to understand the context.



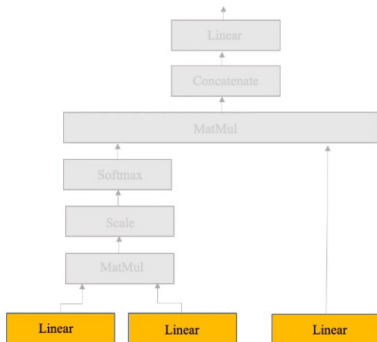
Multi-Head Attention



Multi-Head Attention

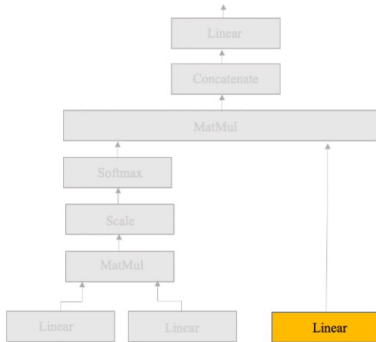


Multi-Head Attention



Multi-Head Attention

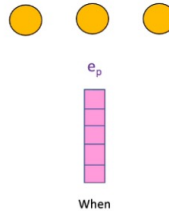
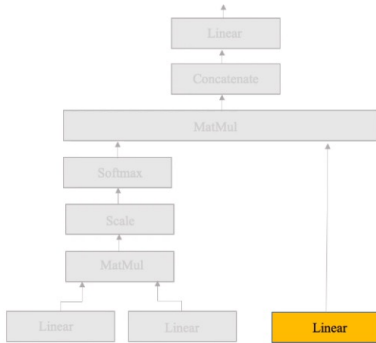
Linear Layer



- i) Mapping inputs onto the outputs
- ii) Changing matrix/vector dimensions

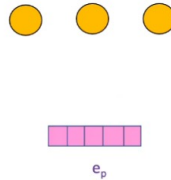
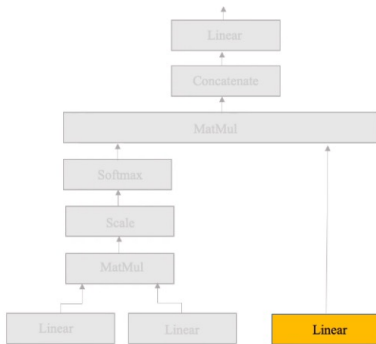
Multi-Head Attention

Linear Layer



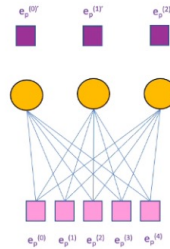
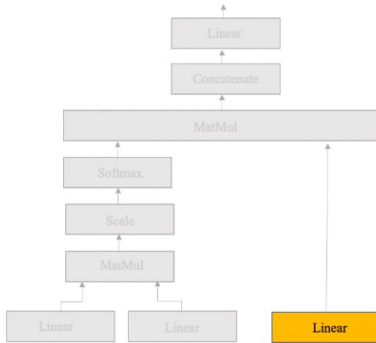
Multi-Head Attention

Linear Layer



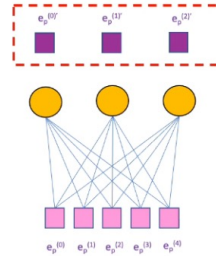
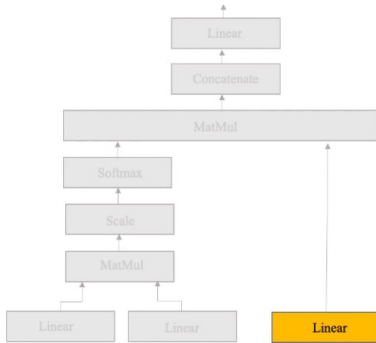
Multi-Head Attention

Linear Layer



Multi-Head Attention

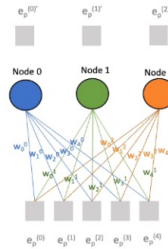
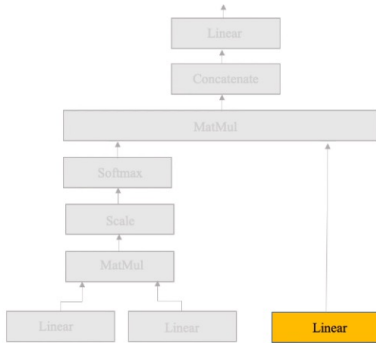
Linear Layer



Multi-Head Attention

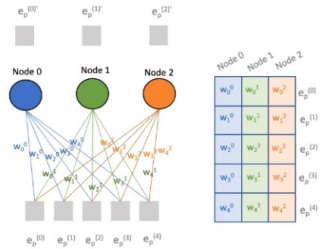
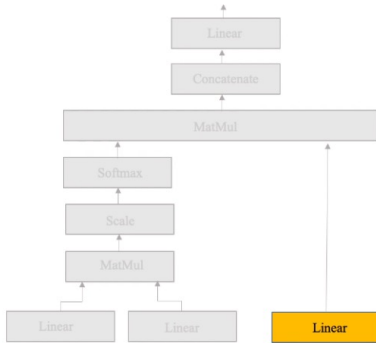
Linear Layer

$$W_0^0 = 0.6$$

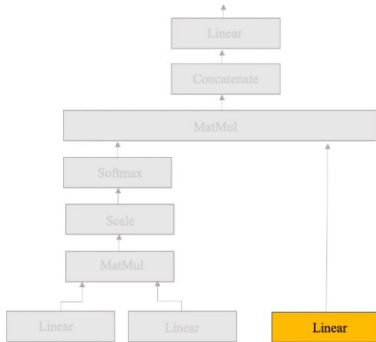


Multi-Head Attention

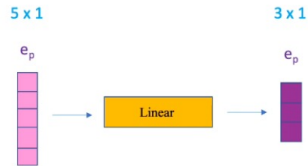
Linear Layer



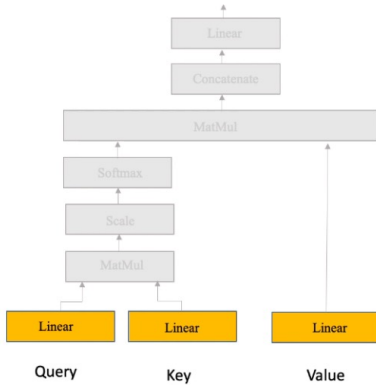
Multi-Head Attention

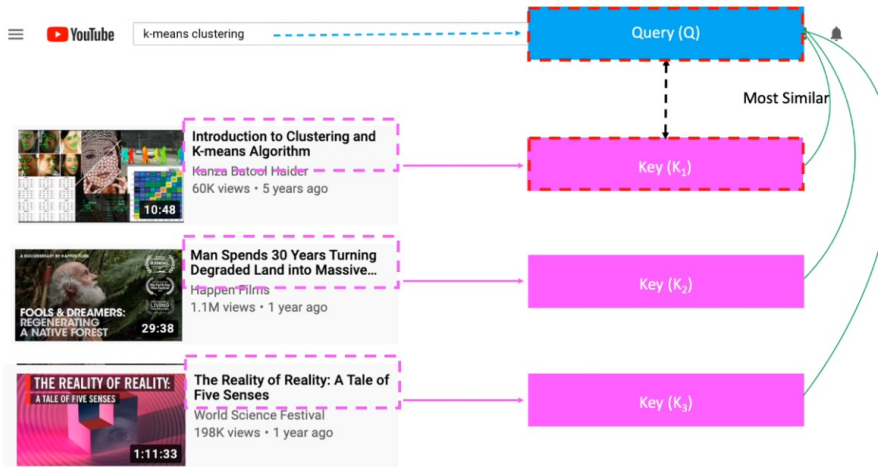


Linear Layer



Multi-Head Attention







Introduction to Clustering and K-means Algorithm

Kanza Batool Haider

60K views • 5 years ago

10:48

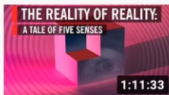


Man Spends 30 Years Turning Degraded Land into Massive...

Happen Films

1.1M views • 1 year ago

29:38



The Reality of Reality: A Tale of Five Senses

World Science Festival

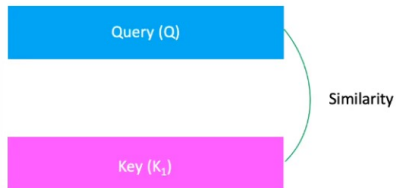
198K views • 1 year ago

1:11:33

Key (K_1)

Value (V_1)

Similarity?



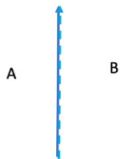
Cosine Similarity

+1 to **-1**

Most similar

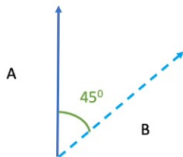
Most dissimilar

Cosine Similarity



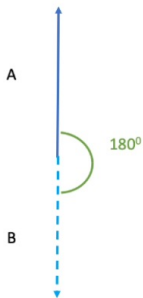
$$\text{Cos}(0) = +1$$

Cosine Similarity



$$\cos(45) = +0.7$$

Cosine Similarity



$$\text{Cos}(180) = -1.0$$

Cosine Similarity

$$\text{Cos}(A, B) = \frac{A \cdot B}{|A| |B|}$$

Similarity b/w Vectors

$$\text{similarity}(A, B) = \frac{A \cdot B}{\text{scaling}}$$

Similarity b/w Matrices

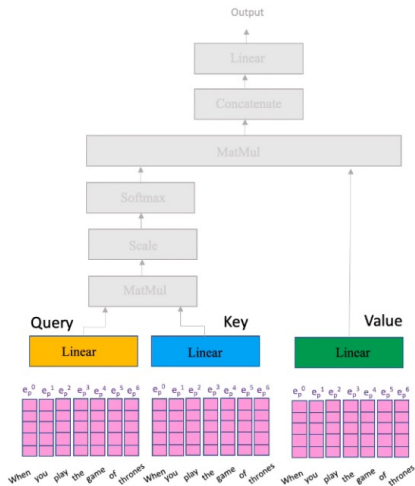
$$\text{similarity}(A, B) = \frac{A \cdot B^T}{\text{scaling}}$$

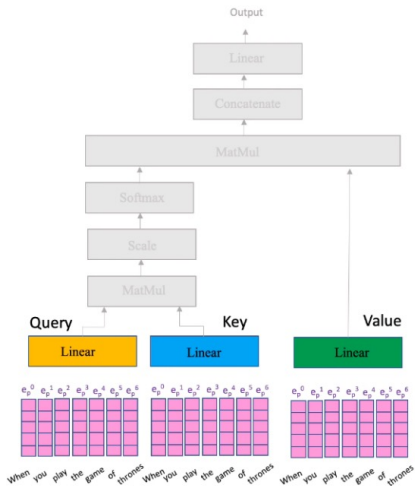
Similarity b/w Query and Key

$$\text{similarity}(\mathbf{Q}, \mathbf{K}) = \frac{\mathbf{Q} \cdot \mathbf{K}^T}{\text{scaling}}$$

Multi-Head Attention

Self-Attention

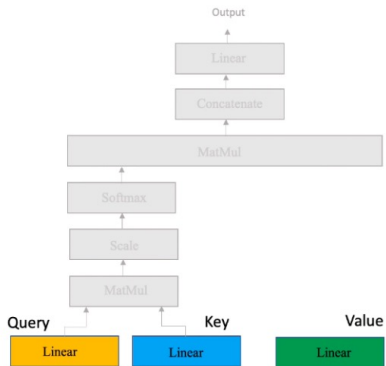




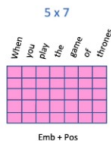
Multi-Head Attention

Self-Attention





Multi-Head Attention



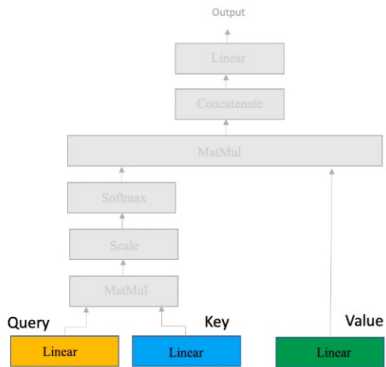
Emb + Pos



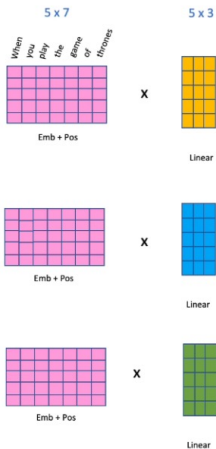
Emb + Pos

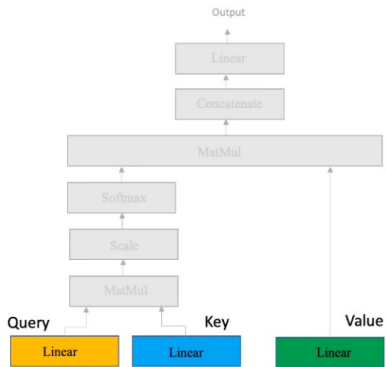


Emb + Pos

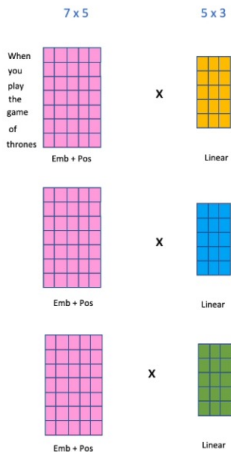


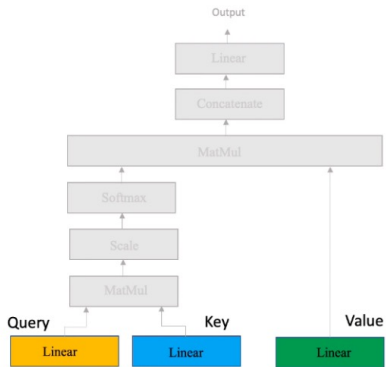
Multi-Head Attention



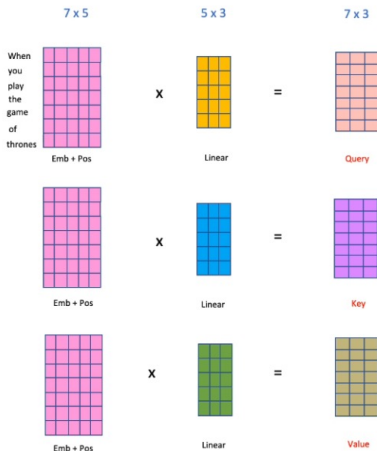


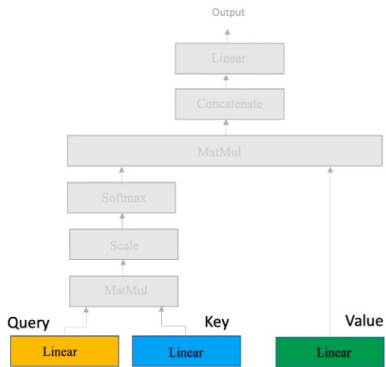
Multi-Head Attention



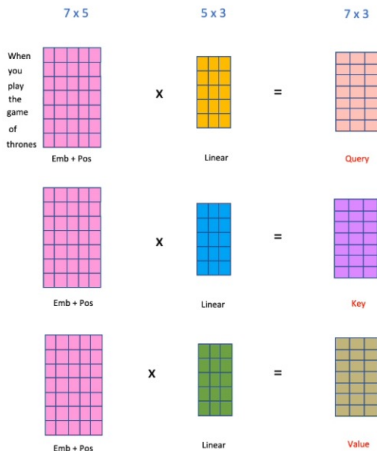


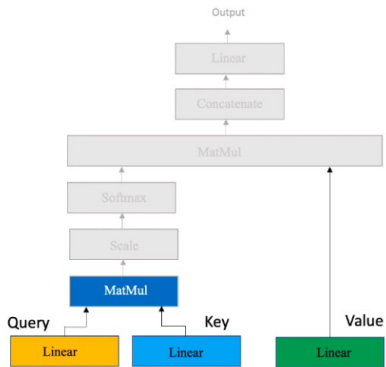
Multi-Head Attention



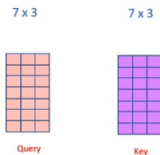


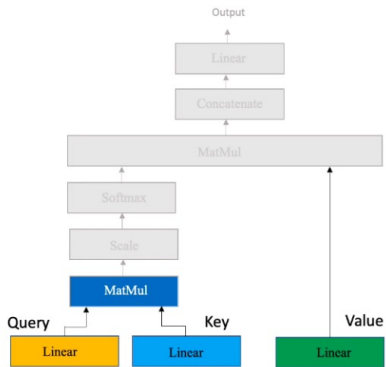
Multi-Head Attention



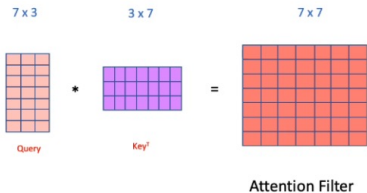


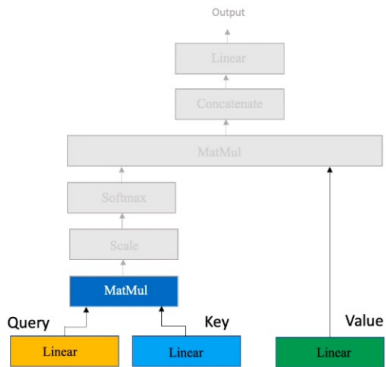
Multi-Head Attention



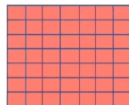


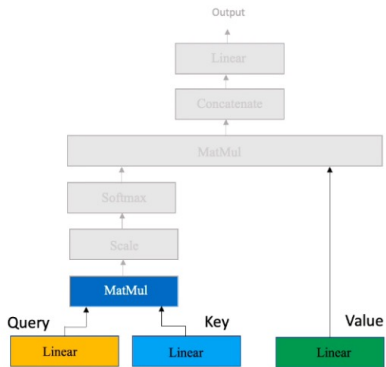
Multi-Head Attention





Multi-Head Attention



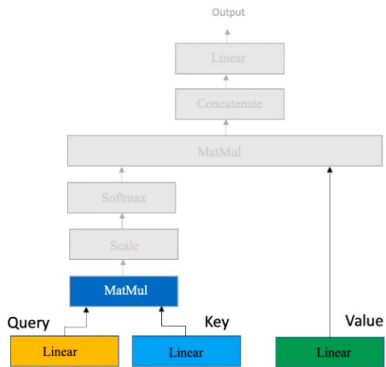


Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	89	48	41	36	35	40	19
you	67	91	11	92	17	99	11
play	91	10	11	11	12	41	98
the	11	96	28	12	98	11	00
game	76	11	91	24	12	12	12
of	11	29	77	78	22	93	13
thrones	11	87	12	12	13	98	19

Initially

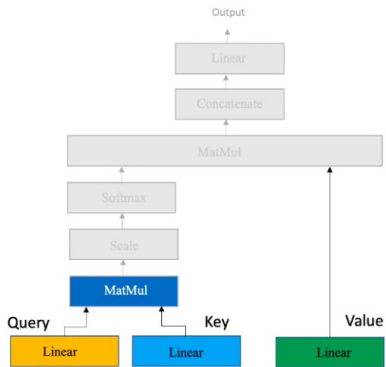


Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	89	20	41	10	55	78	59
you	90	98	81	22	87	15	32
play	29	81	95	10	90	30	92
the	10	22	67	12	88	40	89
game	22	70	90	56	98	44	80
of	10	15	30	40	44	44	59
thrones	59	72	92	90	13	59	99

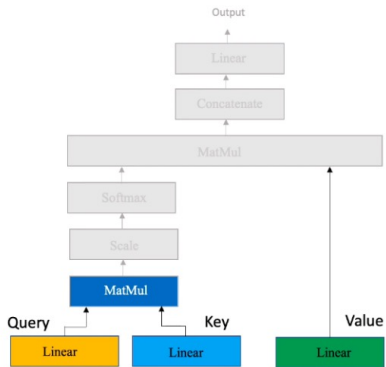
After training



Multi-Head Attention

7 x 7

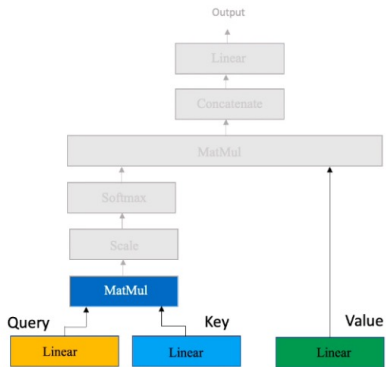
	When	you	play	the	game	of	thrones
When	89	20	41	10	55	10	59
you	20	90	81	22	70	15	72
play	41	81	95	10	90	30	92
the	10	22	10	92	88	40	89
game	55	70	90	88	98	44	87
of	10	15	30	40	44	85	59
thrones	59	72	92	90	95	59	99



Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	89	20	41	10	55	10	59
you	20	90	81	22	70	15	72
play	41	81	95	10	90	30	92
the	10	22	10	92	88	40	89
game	55	70	90	88	98	44	87
of	10	15	30	40	44	85	59
thrones	59	72	92	90	95	59	99

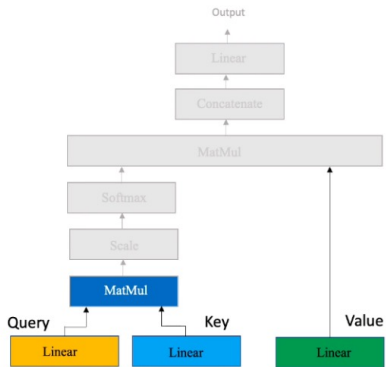


Multi-Head Attention

7 x 7

When you play the **game** of thrones

When	89	20	41	10	55	10	59
you	20	90	81	22	70	15	72
play	41	81	95	10	90	30	92
the	10	22	10	92	88	40	89
game	55	70	90	88	98	44	87
of	10	15	30	40	44	85	59
thrones	59	72	92	90	95	59	99

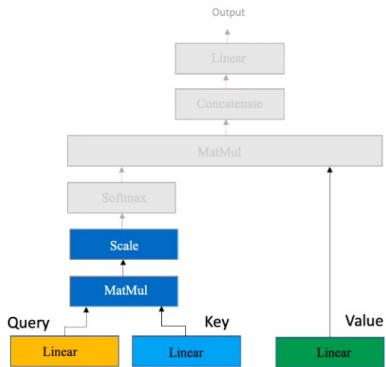


Multi-Head Attention

7 x 7

When you play the game of thrones

When	89	20	41	10	55	10	59
you	20	90	81	22	70	15	72
play	41	81	95	10	90	30	92
the	10	22	10	92	88	40	89
game	55	70	90	88	98	44	87
of	10	15	30	40	44	85	59
thrones	59	72	92	90	95	59	99

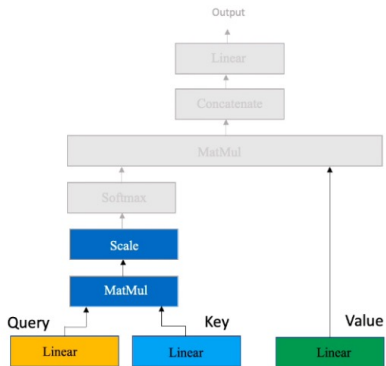


Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	89	20	41	10	55	10	59
you	20	90	81	22	70	15	72
play	41	81	95	10	90	30	92
the	10	22	10	92	88	40	89
game	55	70	90	88	98	44	87
of	10	15	30	40	44	85	59
thrones	59	72	92	90	95	59	99

$$\sqrt{d_k}$$

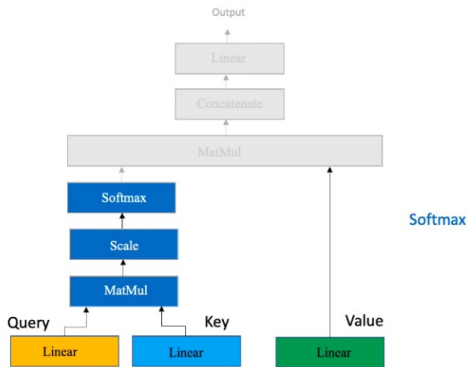


Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	89	20	41	10	55	10	59
you	20	90	81	22	70	15	72
play	41	81	95	10	90	30	92
the	10	22	10	92	88	40	89
game	55	70	90	88	98	44	87
of	10	15	30	40	44	85	59
thrones	59	72	92	90	95	59	99

$\sqrt{7}$

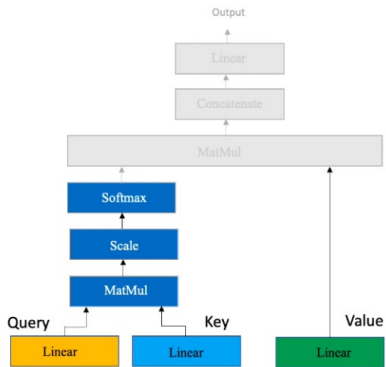


Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	33.6	7.6	15.5	3.8	20.8	3.8	22.3
you	7.6	34.0	30.6	8.3	26.5	5.7	27.2
play	15.5	30.6	35.9	3.8	34.0	11.3	34.8
the	3.8	8.3	3.8	34.8	33.3	15.1	33.6
game	20.8	26.5	34.0	33.3	37.0	16.6	35.9
of	3.8	5.7	11.3	15.1	16.6	32.1	22.3
thrones	22.3	27.2	34.8	34.0	35.9	22.3	37.4

$$\text{softmax}(x_i) = \frac{\exp(x_i)}{\sum_j \exp(x_j)}$$



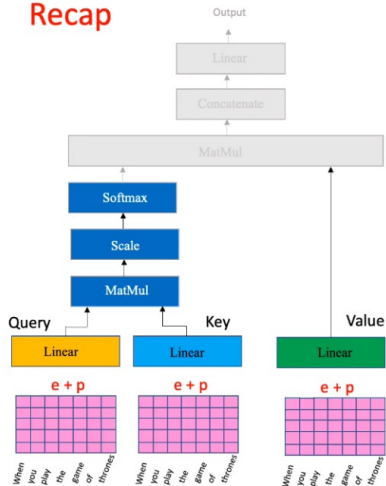
Multi-Head Attention

7 x 7

	When	you	play	the	game	of	thrones
When	1.00	0.00	0.00	0.00	0.00	0.00	0.00
you	0.00	0.97	0.03	0.00	0.00	0.00	0.00
play	0.00	0.00	0.68	0.00	0.10	0.00	0.22
the	0.00	0.00	0.00	0.65	0.14	0.00	0.21
game	0.00	0.00	0.03	0.02	0.72	0.00	0.23
of	0.00	0.00	0.00	0.00	0.00	1.00	0.00
thrones	0.00	0.00	0.05	0.03	0.17	0.00	0.75

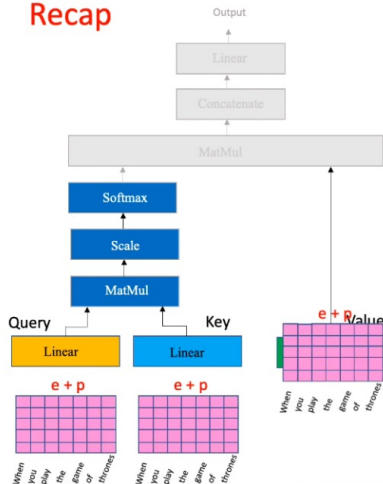
Attention Filter

Recap

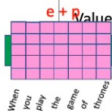


Multi-Head Attention

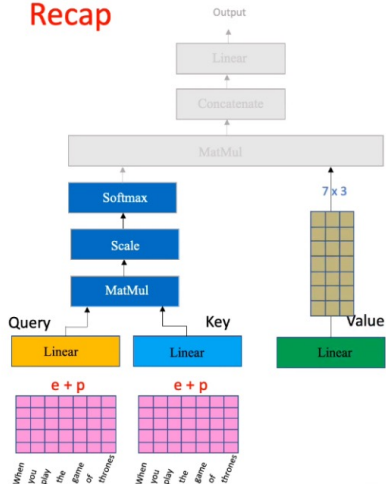
Recap



Multi-Head Attention

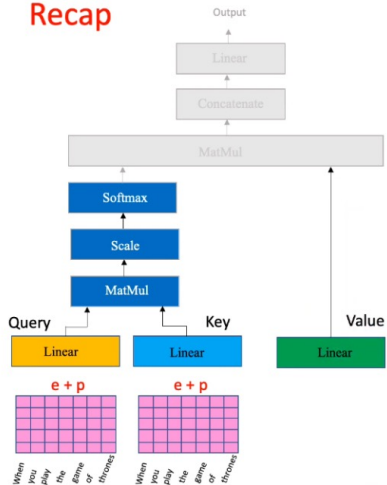


Recap



Multi-Head Attention

Recap

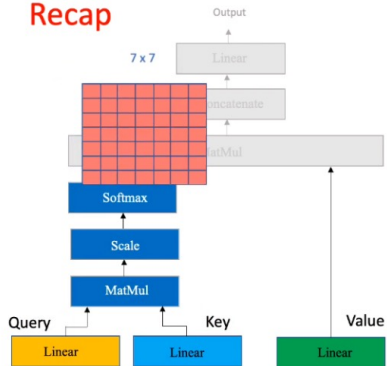


Multi-Head Attention

7 × 3



Recap

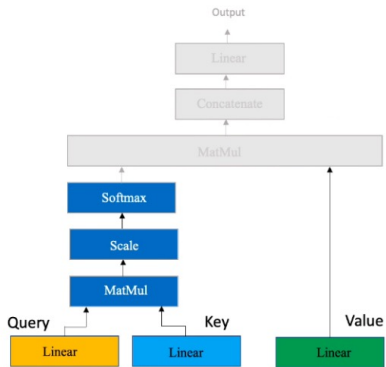


Multi-Head Attention

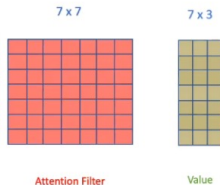
7 x 3

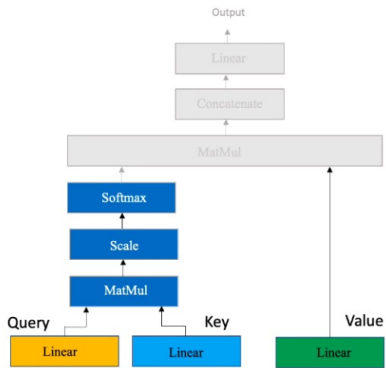


Value

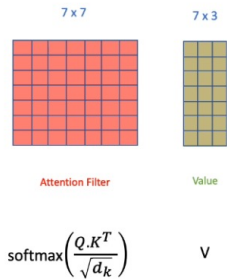


Multi-Head Attention





Multi-Head Attention

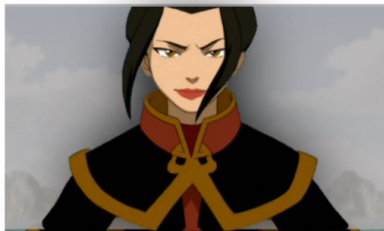


Intuition



Princess Azula of the Fire Nation

Intuition



Intuition

Attention Filter

Original Image

Filtered Image

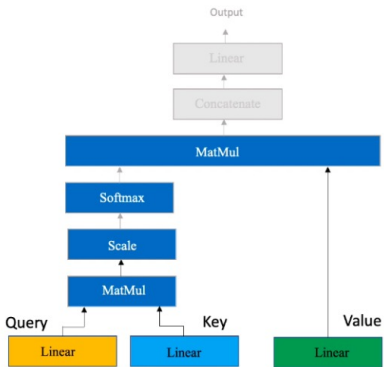


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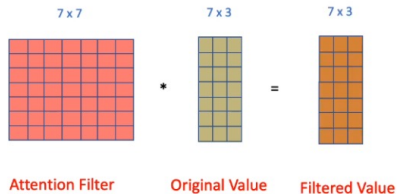


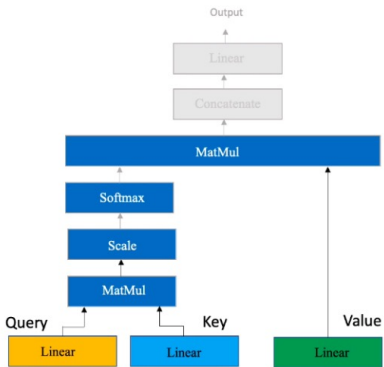
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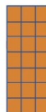
Multi-Head Attention





Multi-Head Attention

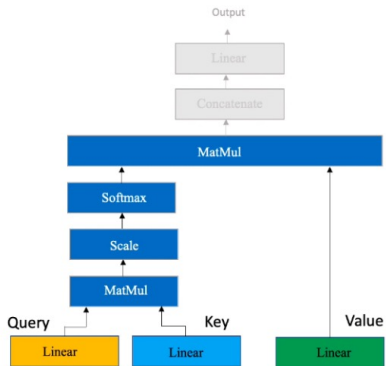
7 x 3



Filtered Value

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \cdot V$$

Multi-Head Attention

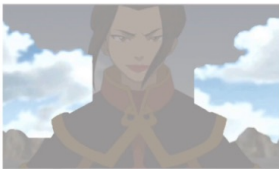


Intuition for Multi-head Attention

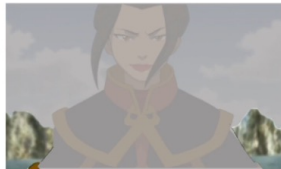
Attention Filter 1



Attention Filter 2

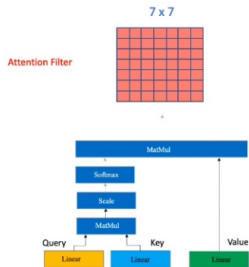


Attention Filter 3

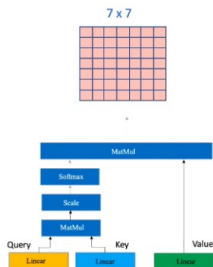


Multi-Head Attention

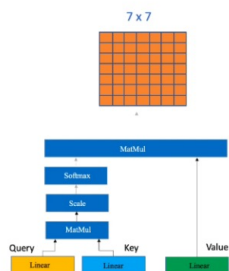
Attention Head 1



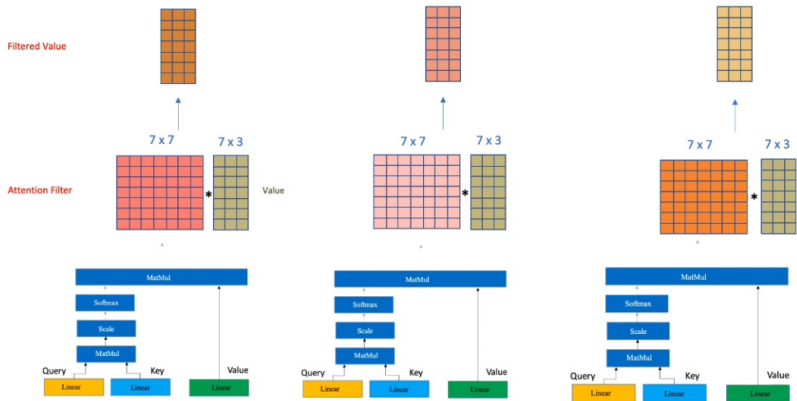
Attention Head 2



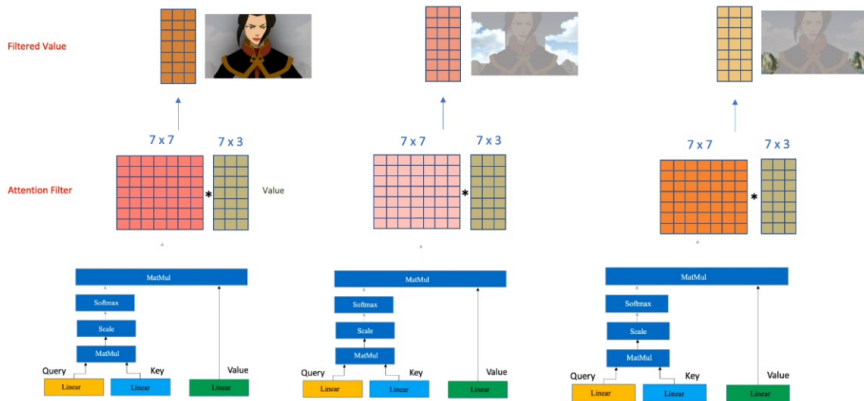
Attention Head 3



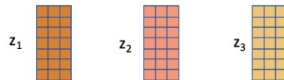
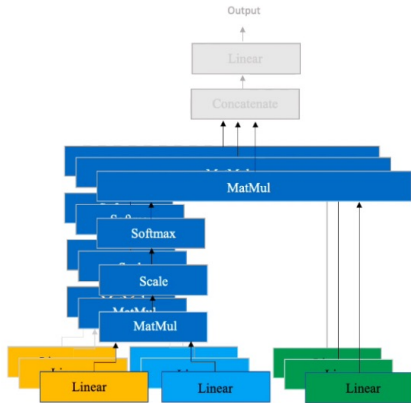
Multi-Head Attention



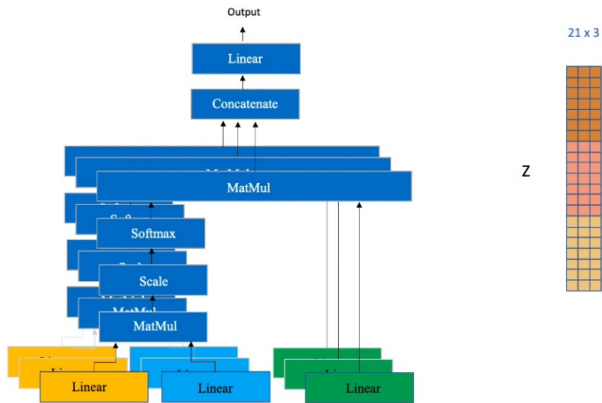
Multi-Head Attention



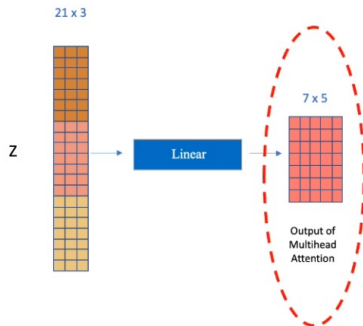
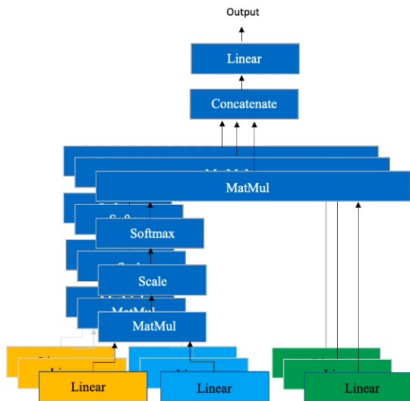
Multi-Head Attention

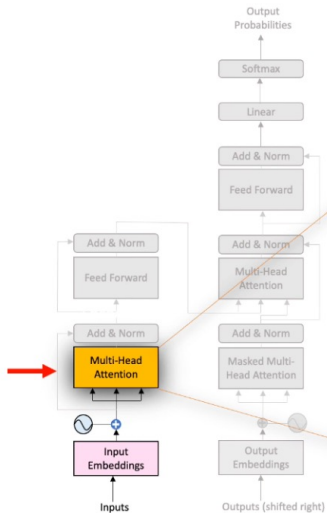


Multi-Head Attention



Multi-Head Attention





Multi-Head Attention

